

Q1:- In a 3D game engine, a surface is defined by the fn $f(x,y) = \sin(x)\cos(y)$. A light source is located at a pt that creates a light vector $L = \langle 0, 0, 1 \rangle$ (pointing straight down).

(a) Find the normal line eq at pt $(\frac{\pi}{4}, \frac{\pi}{4})$ to determine how light reflects off the 'micro-geometry'.

(b) Find the tangent plane at this pt, which represents the flat 'shading proxy' for that pixel.

Sol $g(x,y,z) = \sin(x)\cos(y) - z = 0$

$$g_x = \cos(x)\cos(y)$$

$$g_x|_{(\frac{\pi}{4}, \frac{\pi}{4})} = \cos(\frac{\pi}{4})\cos(\frac{\pi}{4}) = 0.5$$

$$g_y = -\sin(x)\sin(y)$$

$$g_y|_{(\frac{\pi}{4}, \frac{\pi}{4})} = -\sin(\frac{\pi}{4})\sin(\frac{\pi}{4}) = -0.5$$

$$g_z = -1$$

$$\text{Normal vector} = \vec{n} = \langle 0.5, -0.5, -1 \rangle$$

$$z = \sin(\frac{\pi}{4})\cos(\frac{\pi}{4}) = 0.5$$

$$P(\frac{\pi}{4}, \frac{\pi}{4}, 0.5)$$

Normal line

$$x = \frac{\pi}{4} + 0.5t$$

$$y = \frac{\pi}{4} - 0.5t$$

$$z = 0.5 - t$$

Tangent Plane

$$0.5(x - \frac{\pi}{4}) - 0.5(y - \frac{\pi}{4}) - 1(z - 0.5) = 0$$

$$0.5x - 0.5y - z = -0.5$$

Q2:- A laser beam travels along the line

$$x = 3 + 2t, \quad y = 1 - t, \quad z = 4 + t$$

A wall is modeled by plane

$$2x - y + z - 8 = 0$$

- (a) Determine whether the laser hits the wall or travels parallel to it.
 (b) If it hits, find the exact collision pt
 (c) If //, compute shortest distance.

sol Direction vector = $\vec{v} = \langle 2, -1, 1 \rangle$

Plane normal = $\vec{n} = \langle 2, -1, 1 \rangle$

They are // $\rightarrow$ laser is $\perp$ to plane
 So it must intersect

Substitute into plane

$$2(3+2t) - (1-t) + (4+t) - 8 = 0$$

$$6t + 1 = 0$$

$$t = -\frac{1}{6}$$

Collision pt : $\left(3 - \frac{2}{6}, 1 + \frac{1}{6}, 4 - \frac{1}{6} \right)$
 $\left(\frac{16}{6}, \frac{7}{6}, \frac{23}{6} \right)$

Q3:- A drone hovers above a landing pad with surface
 $z = 5 - x^2 - y^2$

At pt $P(1,1)$, find:

- (a) Eq of tangent plane
 (b) Parametric eq of normal line to surface at P .
 (c) Distance from drone at $D(1,1,6)$ along normal line to landing surface.

sol $f_x = -2x$, $f_y = -2y$

At $(1,1)$

$$f_x = -2$$
 , $f_y = -2$

Tangent Plane $z - z_0 = f_x(x - x_0) + f_y(y - y_0)$

$$z - 3 = -2(x - 1) - 2(y - 1)$$

$$Z = -2x - 2y + 7$$

Normal vector : $(f_x, f_y, -1)$
 $(-2, -2, -1)$

For parametric line, can use $(2, 2, 1)$

$$x = 1 + 2t, y = 1 + 2t, z = 3 + t$$

Distance from drone $(1, 1, 6)$ to surface along normal

$$z = 3 + t$$

$$6 = 3 + t$$

$$t = 3$$

Distance = 3 units

Q4:- A ray hits surface $z = x^2 + y^2$ at pt $P(1, 1, 2)$.
 Incoming ray direction is $\vec{v} = \langle 1, -1, -2 \rangle$

(a) Find tangent plane at P

(b) Surface normal at P

(c) check if incoming ray is $\perp$ to surface

sol $\vec{\nabla} F = \langle 2x, 2y, -1 \rangle$

$$\vec{\nabla} F|_{(1,1,2)} = \langle 2, 2, -1 \rangle$$

Tangent Plane

$$z - 2 = 2(x - 1) + 2(y - 1)$$

$$z = 2x + 2y - 2$$

check perpendicularity

$$\vec{v} \cdot \vec{n} = 1 \times 2 + (-1) \times 2 + (-2) \times (-1) \\ = 2$$

Ray is not $\perp$ to surface

Q5:- In a neural network, a loss fn is modeled locally as $f(x, y) = x^2 e^y$. A researcher is at pt $P(2, 0)$ & decides to take a small step of size $s = 0.1$ in the direction of

$\vec{V} = \langle 3, -4 \rangle$. Estimate the change in loss ftn.

sol

$$f_x = 2xe^y$$

$$f_x(2,0) = 4$$

$$f_y = x^2 e^y$$

$$f_y(2,0) = 4$$

$$\vec{\nabla} f(2,0) = 4\hat{i} + 4\hat{j}$$

$$|\vec{V}| = \sqrt{(3)^2 + (-4)^2} = 5$$

$$\vec{U} = \langle 0.6, -0.8 \rangle$$

$$df = (\vec{\nabla} f(2,0) \cdot \vec{U}) ds$$

$$= (-0.8)(0.1)$$

$$= -0.08$$

The loss ftn decreases by approximately 0.08 units.

Q6:- In computer graphics, light intensity I on a surface is $I(x,y,z) = \frac{100}{x^2+y^2+z^2}$. An observer at $(1,2,2)$ moves a tiny distance $ds = 0.05$ towards the pt $(2,1,1)$. Estimate change in light intensity.

sol

$$I_x \Big|_{(1,2,2)} = \frac{-200x}{(x^2+y^2+z^2)^2} = -2.47$$

$$I_y \Big|_{(1,2,2)} = \frac{-200y}{(x^2+y^2+z^2)^2} = -4.94$$

$$I_z \Big|_{(1,2,2)} = \frac{-200z}{(x^2+y^2+z^2)^2} = -4.94$$

$$\vec{V} = \langle 1, 1, -1 \rangle$$

$$\vec{U} = \frac{\vec{V}}{|\vec{V}|} = \frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} - \frac{1}{\sqrt{3}}\hat{k}$$

$$dI = (\vec{\nabla} I(1,2,2) \cdot \vec{U}) ds$$

$$dI = (-1.426)(0.05)$$

$$dI = -0.0713$$

The intensity decreases by approximately 0.07 units.